

Trade-offs, Comparative Advantage, and the Market System

ECON 101 – Principles of Macroeconomics

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Spring 2026

- 2.1 Production Possibilities Frontiers and Opportunity Costs
- 2.2 Comparative Advantage and Trade
- 2.3 The Market System

Scarcity and Trade-offs

- People continually face decisions about how best to use their **scarce** resources.
- **Scarcity**: a situation in which unlimited wants exceed the limited resources available to fulfill those wants.
- Scarcity requires **trade-offs**: to get more of one thing, society must accept less of another.
- Economics gives us tools to help make these trade-offs wisely.

Canadian Example: Alberta's Energy Trade-off

- Alberta has limited land, capital, and labour.
- If Alberta devotes more resources to **oil sands extraction**, fewer resources are available for **renewable energy** development (wind/solar).
- This trade-off is real: the Trans Mountain pipeline expansion (completed 2024) required massive capital that could have been invested elsewhere.
- **Source:** Natural Resources Canada; Statistics Canada Table 25-10-0063-01.

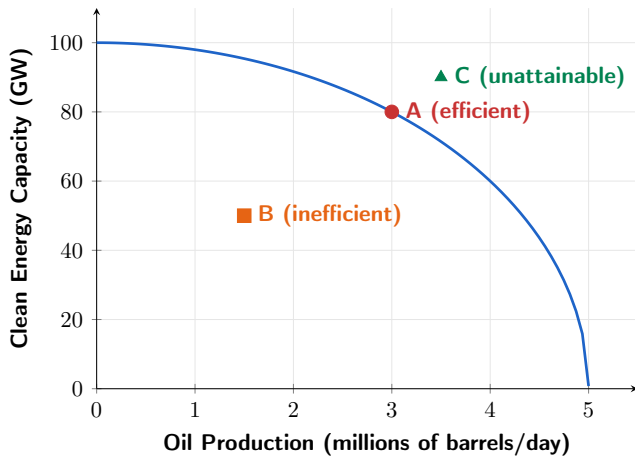
2.1 Learning Objective

Goal

Use a production possibilities frontier to analyze opportunity costs and trade-offs.

- A **production possibilities frontier (PPF)** is a curve showing the maximum attainable combinations of two goods that can be produced with available resources and current technology.
- The PPF is a **positive** tool: it shows what *is* possible, not what *should* be produced.

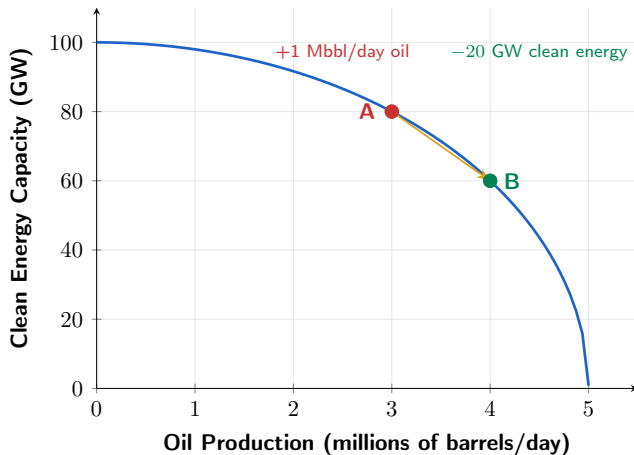
Figure 2.1 – Canada's PPF: Oil vs. Clean Energy (1 of 2)



Interpreting the PPF: Canada's Energy Sector

- Canada's economy can produce combinations of **oil output** and **clean energy capacity**.
- **Point A** (on the PPF): productively efficient — all resources are fully used.
- **Point B** (inside the PPF): attainable but **inefficient** — idle workers or unused capital (e.g., pandemic-era shutdowns, 2020).
- **Point C** (outside the PPF): currently **unattainable** with existing resources and technology.
- **Data context:** Canada produced ≈ 4.9 million barrels/day of oil in 2024 and had ≈ 100 GW of total electricity capacity (Statistics Canada / NRCan).

Figure 2.1 – Moving Along the PPF: Opportunity Cost (2 of 2)

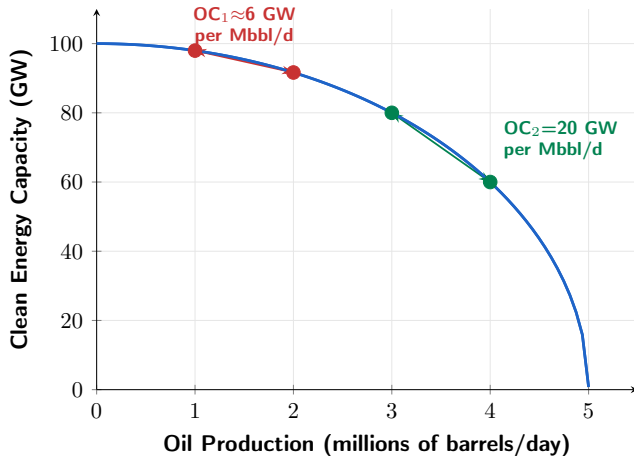


Moving from A→B: gaining 1 Mbbl/day of oil costs **20 GW of clean energy capacity** — that is the opportunity cost.

Opportunity Cost on the PPF

- Moving from point A to point B:
 - Canada produces **1 million more barrels/day** of oil.
 - To do so, it must give up **20 GW** of clean energy capacity.
- The **opportunity cost** of that extra oil is the clean energy capacity sacrificed.
- **Opportunity cost**: the highest-valued alternative that must be given up to engage in an activity.
- Real-world context: the federal government's Clean Electricity Regulations (2024) explicitly recognize this trade-off.

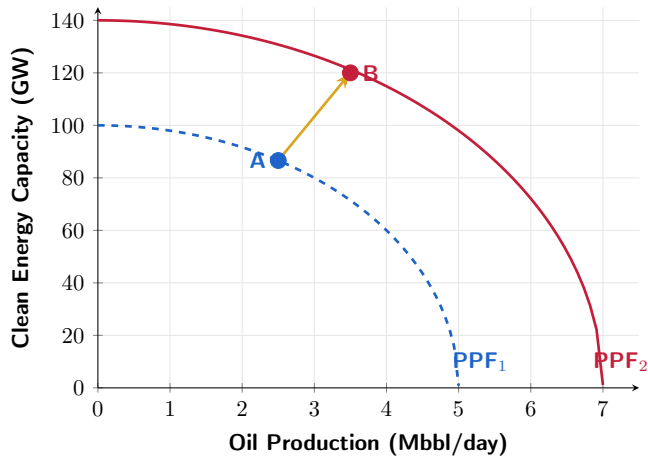
Figure 2.2 – Increasing Marginal Opportunity Costs



Why Opportunity Costs Often Increase

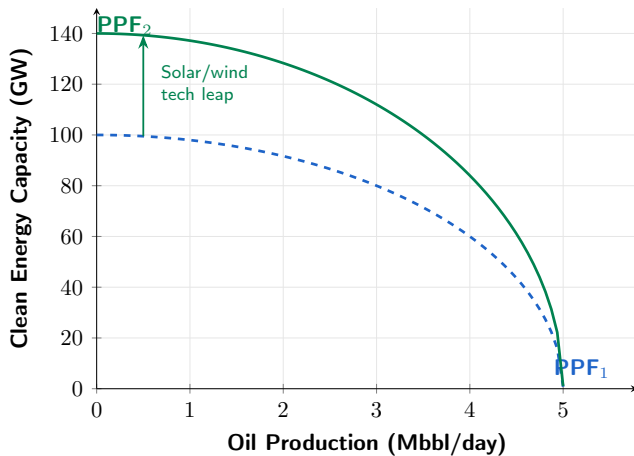
- Some resources are better suited to oil production, others to clean energy.
- The *first* resources shifted from oil to clean energy are the ones *most easily* repurposed.
- As more resources are diverted, the extra output gained falls — giving the PPF its **bowed-out shape**.
- **Canadian example:** Shallow conventional oil fields in Saskatchewan are relatively easy to repurpose; deep oil sands operations in Alberta are highly specialized — shifting them costs far more clean energy capacity.

Figure 2.3a – Economic Growth: Outward Shift of the PPF



- As more resources become available, the economy moves from A to B — producing **more of both** goods.
- An **outward shift** of the PPF represents **economic growth**.
- **Economic growth**: the ability of the economy to increase the production of goods and services.
- **Canadian data**: Canada's real GDP grew by about 2.4% per year on average 2000–2023 (Statistics Canada, CANSIM 36-10-0104-01). Each year, the PPF expanded outward.
- Macro link: long-run growth allows higher living standards without giving up other goods.

Figure 2.3b – Technological Change in One Sector



Technological Change in One Industry

- Suppose technology improves in **clean energy only** (e.g., cheaper solar panels, better battery storage).
- The maximum clean energy capacity increases, but the oil production frontier is unchanged.
- Previously unattainable combinations become attainable.
- **Canadian example:** The cost of utility-scale solar in Canada dropped by $\approx 89\%$ from 2010 to 2023, allowing the clean energy PPF frontier to expand significantly (IRENA / NRCan data).

2.2 Learning Objective

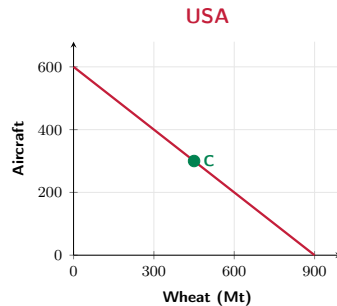
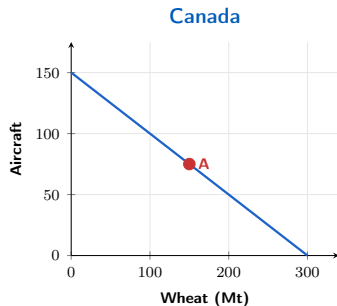
Goal

Understand comparative advantage and explain how it serves as the basis for trade.

Canada–USA Trade Example

- Canada and the USA each have limited land, labour, and capital.
- Suppose both can produce **wheat** and **aircraft**.
- **In one year**, with all resources devoted to one good:
 - **Canada:** 300 Mt of wheat *or* 150 aircraft.
 - **USA:** 900 Mt of wheat *or* 600 aircraft.
- The USA is **absolutely** better at producing both.
- Yet both countries can **still gain** from trade — the key is **comparative advantage**.

Figure 2.4 – Canada & USA PPFs Without Trade



Without trade, each country consumes only what it produces (points A and C).

Autarky (No Trade) Outcomes

- Without trade, each country consumes only what it produces.
- **Canada (Point A):** produces and consumes 150 Mt wheat and 75 aircraft.
- **USA (Point C):** produces and consumes 450 Mt wheat and 300 aircraft.
- Both are on their own PPFs, but *neither* is at the best possible consumption point once trade is allowed.

Specialization and Gains from Trade

- **Trade:** the act of buying and selling across borders.
- The USA is absolutely better at producing both wheat and aircraft.
- But each country should specialize in what it is **relatively** better at.
- The key concept: **comparative advantage** — produce where your *opportunity cost* is lower.

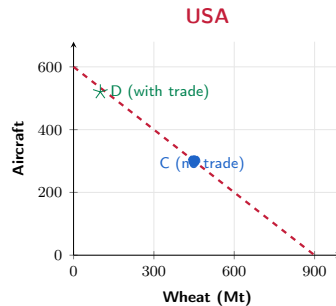
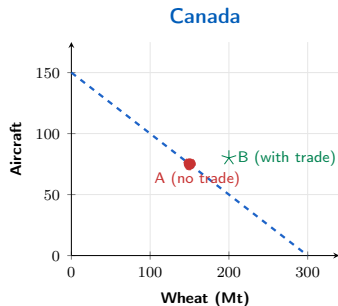
Table 2.1 – Opportunity Costs of Wheat and Aircraft

	Canada	USA
OC of 1 Mt wheat	$\frac{150}{300} = 0.5$ aircraft	$\frac{600}{900} = 0.67$ aircraft
OC of 1 aircraft	$\frac{300}{150} = 2$ Mt wheat	$\frac{900}{600} = 1.5$ Mt wheat
Comparative advantage in:	Wheat	Aircraft

Key insight

Canada gives up *less* to produce wheat; the USA gives up *less* to produce aircraft.

Figure 2.5 – Gains from Trade



Both countries reach consumption points **outside** their own PPFs — gains from trade!

Table 2.2 – Summary of the Gains from Trade

	Canada		USA	
	Wheat (Mt)	Aircraft	Wheat (Mt)	Aircraft
Without trade	150	75	450	300
With trade	200	80	100	520
Gain	+50	+5	–	+220

Key lesson

Both countries consume *more* with trade — even though the USA is absolutely better at producing everything.

Explaining the Gains from Trade

- The USA has an **absolute advantage** in both wheat and aircraft.
- Canada has a **comparative advantage** in wheat (lower opportunity cost).
- The USA has a **comparative advantage** in aircraft.
- **Absolute advantage**: the ability to produce more of a good using the same resources.
- **Comparative advantage**: the ability to produce a good at a *lower opportunity cost*.
- Real context: Canada exports $\approx$ \$12B of wheat/canola annually; the USA exports $\approx$ \$130B of aerospace products (Statistics Canada; U.S. BEA, 2023).

Canada's Comparative Advantages Today

- **Natural resources:** potash, lumber, oil, canola, uranium — Canada is a low-opportunity-cost producer.
- **Financial services:** Canadian banks are globally competitive.
- **AI & technology:** the “Toronto–Waterloo corridor” and Montreal AI cluster reflect a comparative advantage in AI research.
- **High opportunity cost goods:** electronics assembly, consumer goods — Canada imports these from countries with comparative advantages there.
- **Source:** Statistics Canada, International Trade Division; Bank of Canada Business Outlook Survey, 2024.

2.3 Learning Objective

Goal

Explain the basic idea of how a market system works.

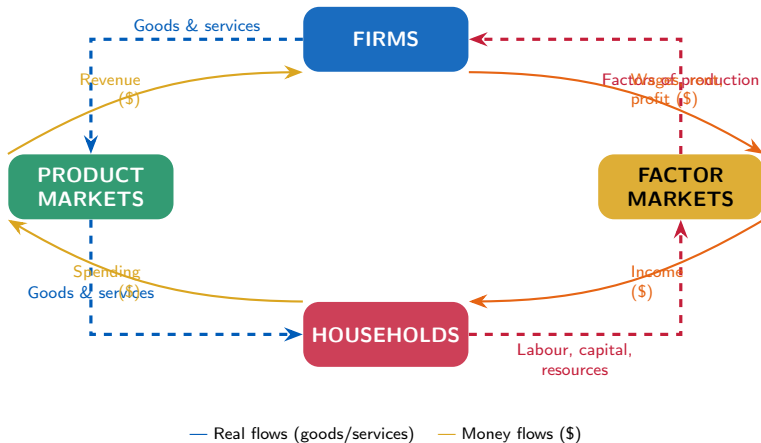
- A **market** is a group of buyers and sellers of a good or service, and the institution or arrangement by which they come together to trade.
- Two key groups participate in the modern economy:
 - **Households**
 - **Firms**
- **Canadian context:** Canada has ≈ 15 million households and ≈ 1.2 million employer businesses (Statistics Canada, 2024).

- **Households** consist of individuals who provide **factors of production**:
 - labour, capital, natural resources, and entrepreneurial ability.
- Households receive payments for these factors by selling them to firms in **factor markets**.
- **Firms** supply goods and services to **product markets**; households buy these from firms.

The Four Factors of Production

- **Labour:** all types of work — from baristas at Tim Hortons to executives at RBC.
- **Capital:** machines, buildings, computers — e.g., pipelines, data centres, manufacturing plants.
- **Natural resources:** land, water, oil sands, forests, mineral deposits.
- **Entrepreneurial ability:** organizing the other factors to create value — e.g., founding a startup on the Toronto–Waterloo corridor.
- **Source:** Statistics Canada, Labour Force Survey; Bank of Canada, Monetary Policy Report (April 2025).

Figure 2.6 – The Circular-Flow Diagram



Circular-Flow Diagram: Basic Idea

- The **circular-flow diagram** illustrates how participants in markets are linked.
- **Real flows:**
 - Households provide factors of production to firms.
 - Firms provide goods and services to households.
- **Money flows:**
 - Firms pay wages, rent, and profits to households for factors.
 - Households pay for goods and services from firms.
- **Canadian scale:** Total household spending $\approx$ \$1.7 trillion in 2024 (Statistics Canada, National Accounts).

Limits of the Simple Circular-Flow Model

- Like all economic models, the circular-flow diagram is a **simplified** version of reality.
- The basic version excludes:
 - **Government** (taxes, transfers, public services)
 - **Financial system** (Bank of Canada, chartered banks, credit markets)
 - **Foreign sector** (exports, imports, exchange rates)
- We add all three sectors in later chapters to build full macroeconomic models.

- A **free market** has few government restrictions on how goods or services can be produced, sold, or how factors can be employed.
- Countries closer to this benchmark have generally raised living standards faster.
- Adam Smith's "**invisible hand**": individuals pursuing self-interest collectively coordinate economic activity.
- **Canadian context:** Canada ranks among the top 10 in the World Bank's Ease of Doing Business index; free trade agreements (CUSMA, CPTPP) have expanded market access for Canadian producers.

Things to Remember

- The PPF should **never bow inward** — opportunity cost is non-decreasing.
- The PPF tells us what **can** be produced, not what **should** be produced.
- Someone being better at *everything* does **not** eliminate gains from trade.
- The basis for trade is **comparative advantage**, not absolute advantage.
- Free markets raise living standards, but governments must provide a sound legal environment (property rights, contract enforcement) for markets to function.

- PPFs illustrate trade-offs, opportunity cost, economic growth, and technological change.
- **Comparative advantage** explains why Canada, the USA, and other nations gain from specialization and trade.
- The **market system** coordinates decisions of millions of households and firms through product and factor markets.
- The **circular-flow diagram** provides a macro picture of how goods, services, and money move in the economy.
- Free markets and well-functioning institutions are central to long-run improvements in living standards.

Thank you!

Questions?

Data sources: Statistics Canada, Bank of Canada, Natural Resources Canada, NRCan, IRENA